

Studying Loss Functions and the Training Criterion

Implementation in NumPy and empirical study on MNIST

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Advanced Data Analytics Algorithms, Machine Learning
Assessment Task 2

Notebook: https://github.com/MathgeniusTB2/a2-mlp-from-scratch/blob/main/a2_mlp_numpy.ipynb

Task and inputs/outputs (head first)

Task: classify a hand-written digit into one of ten classes.

Training

- Input: 28×28 image of integers in $[0, 255]$, normalised to $[0, 1]$, flattened to $\mathbb{R}^{784}$.
- Target: one-hot vector in $\{0, 1\}^{10}$.

Deployment

- Input: one image, same deterministic preprocessing.
- Output: $p \in [0, 1]^{10}$, $\sum_k p_k = 1$; decision $\arg \max_k p_k$.

Research question: how does the choice of training criterion change what the network learns, and when does minimising the loss fail to serve the task objective?

The five criteria

Same MLP 784-128-128-10 (tanh, 118 282 params); only the loss changes.

- **Cross-entropy:** $-\frac{1}{n} \sum y \log \hat{p}$ (proper, reference).
- **Squared error:** $\frac{1}{n} \sum \frac{1}{2} \|\hat{p} - y\|^2$.
- **Absolute error:** $\frac{1}{n} \sum \|\hat{p} - y\|_1$.
- **Label smoothing:** CE against $(1 - \epsilon)y + \epsilon/K$.
- **Expected cost (mine):** $\frac{1}{n} \sum \hat{p}_k(1 - R_{y,k})$.

My reward matrix

rows = true digit, columns = guess; diagonal 10, visually similar digits get up to 6 (e.g. 4:9 = 6, 3:5 = 5). The criterion is the expected cost of the decision, so a near miss is cheaper.

Gradients: theory to code

Only $\delta^{(L)} = \partial \mathcal{L} / \partial z^{(L)}$ changes between criteria

Criterion	logit gradient
cross-entropy	$\frac{1}{n}(\hat{p} - y)$
squared error	$\frac{1}{n}\hat{p} \odot ((\hat{p} - y) - \sum_k (\hat{p}_k - y_k)\hat{p}_k)$
absolute error	$\frac{1}{n}\hat{p} \odot (\text{sign}(\hat{p} - y) - \sum_k \text{sign}(\hat{p}_k - y_k)\hat{p}_k)$
label smoothing	$\frac{1}{n}(\hat{p} - t)$
expected cost	$\frac{1}{n}\hat{p} \odot (c - \sum_k \hat{p}_k c_k), c = 1 - R_y$

Weight gradient is shared: $\partial \mathcal{L} / \partial W^{(\ell)} = \delta^{(\ell)\top} a^{(\ell-1)}$. The softmax Jacobian gives the $\hat{p}_k(g_k - \sum_j \hat{p}_j g_j)$ form.

Mini-batch Adam, $\eta = 10^{-3}$, 30 epochs

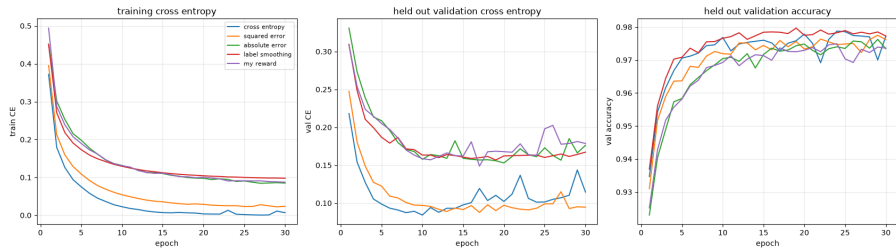
$$\theta \leftarrow \theta - \eta \frac{\hat{m}}{\sqrt{\hat{v}} + \epsilon}$$

- Bias-corrected moments, written from scratch.
- Same optimiser and rate for all five criteria.

Honest protocol

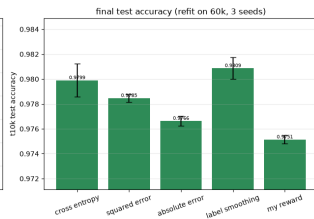
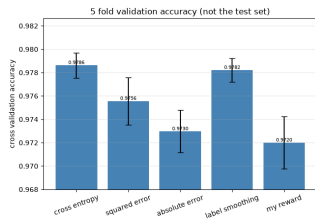
- 5-fold cross-validation on the training set
→ compare criteria (folds are validation, *not* test).
- Refit on all 60 000, score **once** on t10k, over 3 seeds.
- Majority-class baseline: 11.4%.

Training over time



- Train CE falls; held-out validation CE flattens.
- The gap is ordinary overfitting.
- Validation is a held-out fold — never the train set, or the curve would only show fit.

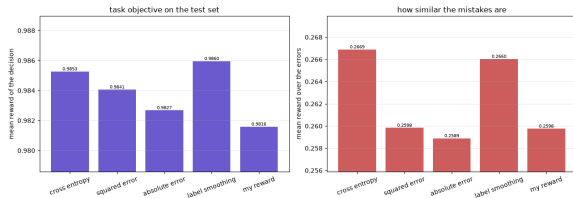
Results



Criterion	CV	Test acc.
label smooth	0.9782	0.9809
cross entropy	0.9786	0.9799
squared error	0.9756	0.9785
absolute error	0.9730	0.9766
my reward	0.9720	0.9751

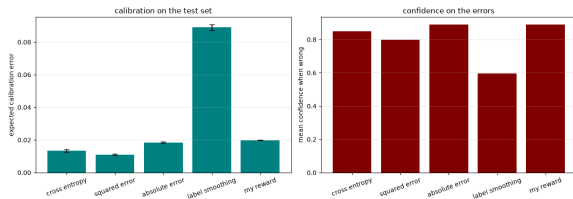
- Accuracy spreads 0.9751–0.9809; choice matters.

Loss versus task objective



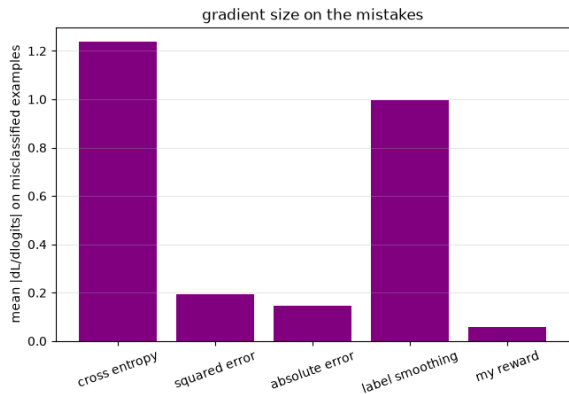
- **Lowest test CE: squared error (0.0826)** — but not the most accurate.
- **Highest accuracy: label smoothing (0.9809)** — but only third-lowest CE (0.1558).
- Loss scores the whole probability vector; accuracy scores only the arg max.
- Label smoothing vs cross-entropy is *not* distinguishable (0.0010 apart, ~ 1 SE).
- My expected-cost criterion is last on accuracy (0.9751).

Calibration



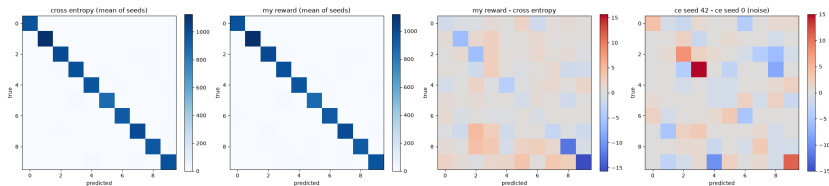
- Measured ECE: squared error 0.011 (best), cross-entropy 0.014, **label smoothing 0.089 (worst)**.
- Label smoothing is the most accurate but the least calibrated (capped confidence, under-confident).
- Mean confidence on errors: 0.60 (label smoothing) vs 0.85 (cross-entropy).

Why the linear criteria fail



- For squared error, absolute error and expected cost, $\delta = \hat{p} \odot (g - \sum_j \hat{p}_j g_j)$: the gradient carries a factor $\hat{p}_k$ and **vanishes when confidently wrong**.
- Cross-entropy's $\delta = \hat{p} - y$ does not.
- Gradient norm on the mistakes: CE 1.24, label smoothing 1.00, squared 0.19, absolute 0.15, mine 0.06.
- Absolute error is hidden-linear: $\|\hat{p} - y\|_1 = 2(1 - \hat{p}_y)$.

Where the mistakes land



- Cross-entropy vs my reward, their difference, and a seed-to-seed noise baseline.
- Largest cell change 15.7 vs noise 15.0: within run-to-run noise.
- The designed cost changed individual errors, not the aggregate; the 3/5/8 and 4/9 structure is the

Limitations and summary

Limitations

- Single hand-designed reward matrix; a data-derived one could differ.
- No finite-difference gradient check; calibrated in-sample; no early stopping.
- Two tanh layers, 30 epochs, 3 seeds: large gaps solid, sub-0.1-point gaps are not.

Summary

- The criterion changes accuracy by 0.6 points under identical training.
- Loss and objective disagree: lowest CE (squared error) is not the most accurate; label smoothing is the most accurate but the worst calibrated.
- The two improper, linear criteria finish last and second-last; their gradient norm on the mistakes is far smaller, consistent with the vanishing-gradient explanation.